

CURRICULUM BOOK

UNDERGRADUATE PROGRAM: BACHELOR OF ENGINEERING

COMPUTER ENGINEERING

Effective from Academic Year: 2026-27

Status: Autonomous College Syllabus Handbook

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ACADEMIC PREAMBLE

As an autonomous engineering college affiliated with the University of Mumbai, Fr. Conceicao Rodrigues College of Engineering is dedicated to engineering excellent learners, innovators, and future technical leaders. Autonomy enables the institution to respond to technological shifts and design responsive curricula that balance theoretical foundations with experiential design, modern tools, and professional ethics.

This curriculum structure represents a unified framework curated in collaboration with academic advisors and leading technical firms. The curriculum emphasizes robust mathematical models, project-based validation, and outcome-oriented professional practice.

Vision of the Department

To be a leading center of excellence in engineering education and research, fostering industry-ready technical innovators, ethically sound professionals, and empathetic citizens.

Mission of the Department

- Deliver standard-aligned engineering education using student-centric pedagogy and laboratory environments.
- Foster research, technical incubation, and industry-institution partnerships.
- Cultivate life-long learning habits, teamwork capabilities, and professional ethical standards.

the 1990s, the number of people in the UK who are employed in the public sector has increased by 1.5 million, from 2.5 million in 1980 to 4 million in 1995 (Department of Health 1996).

There is a growing emphasis on the need to improve the efficiency of the public sector, and to ensure that the public sector is able to deliver the services that are required by the public. This has led to a number of initiatives, including the introduction of competition, the restructuring of public services, and the introduction of new management practices.

One of the main reasons for the need to improve the efficiency of the public sector is the increasing pressure on public resources. The public sector is now responsible for a much larger proportion of the total public expenditure than in the past, and this has led to a need to find ways of reducing costs and improving efficiency.

Another reason for the need to improve the efficiency of the public sector is the increasing demand for public services. The population is ageing, and this has led to a need for more health and social care services. In addition, there is a growing demand for education and training services.

There are a number of ways in which the efficiency of the public sector can be improved. One way is to introduce competition, which can help to reduce costs and improve efficiency. Another way is to restructure public services, so that they are able to deliver services more efficiently.

There are a number of other ways in which the efficiency of the public sector can be improved, including the introduction of new management practices, and the use of information technology. These initiatives are all aimed at reducing costs and improving efficiency, so that the public sector is able to deliver the services that are required by the public.

There is a need to ensure that the public sector is able to deliver the services that are required by the public, and that it is able to do so in a cost-effective way. This is a challenge that the public sector must meet, and it is one that requires the support of the public.

The public sector is a vital part of our society, and it is one that we all rely on. It is important that we ensure that it is able to deliver the services that we need, and that it is able to do so in a cost-effective way. This is a challenge that we must all meet, and it is one that requires the support of the public.

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CURRICULUM ARCHITECTURE AND NEP ALIGNMENT

Course Nomenclature Table

Category	Course Category Name	NEP-2020 Credit Framework Description
BS	Basic Sciences	Fundamental disciplines including Physics, Chemistry, and Advanced Calculus.
ES	Engineering Sciences	Interdisciplinary foundations such as Computing, Graphics, and Electrical Systems.
PC	Professional Core	Core discipline-specific subjects, advanced concepts, and matching practicals.
PE	Professional Electives	Branching streams for specialized expertise such as AI/ML, Security, and IoT.
OE	Open Electives	Multi-disciplinary courses offered across departments for wider learning.
HS	Humanities and Social Sciences	Professional Communication, Business Ethics, IPR, and Management.

SEMESTERWISE CURRICULUM STRUCTURE

Computer Engineering Program: Semester 3

Course Code	Course Name	Teaching Scheme (Contact Hours/Week)			Credits Assigned				Examination Scheme						
		L	T	P	L	T	P	Total	Theory Courses				Practical / Lab		Total Marks
									ISE 1	ISE 2	MSE	ESE	ISE 1	ISE 2	
CSC301	Data Structures and Algorithms	3	1	0	3	1	0	4	20	20	-	60	-	-	100
CSL301	Data Structures Laboratory	0	0	2	0	0	1	1	20	20	-	-	25	25	100
CSC302	Database Management Systems	3	0	0	3	0	0	3	20	20	-	60	-	-	100
CSL302	Database Systems Laboratory	0	0	2	0	0	1	1	20	20	-	-	25	25	100
CSC303	Computer Networks	3	1	0	3	1	0	4	20	20	-	60	-	-	100
CSC304	Operating Systems	3	0	0	3	0	0	3	20	20	-	60	-	-	100
CSE305	Professional Elective: Artificial Intelligence	3	0	0	3	0	0	3	20	20	-	60	-	-	100
CSP306	Mini Project I	0	0	4	0	0	2	2	20	20	-	-	25	25	100
Total		15	2	8	21				-						

Abbreviations: L: Lecture; T: Tutorial; P: Practical/Lab; ISE: In-Semester Evaluation; MSE: Mid-Semester Examination; ESE: End Semester Examination.

Regulatory Ordinance Code: R-2026-COMP-III

the 1990s, the number of people in the UK who are employed in the public sector has increased by 1.5 million, from 2.5 million in 1980 to 4 million in 1995 (Department of Health 1996).

There is a growing emphasis on the need to improve the efficiency of the public sector, and to ensure that the public sector is able to deliver the services that are required by the public. This has led to a number of initiatives, including the introduction of competition, the restructuring of public services, and the introduction of new management practices.

One of the main challenges facing the public sector is the need to improve the efficiency of the services that it provides. This is a complex task, and it requires a number of different approaches. One of the most important is the introduction of competition, which can help to drive down costs and improve the quality of services.

Another important approach is the restructuring of public services, which can help to eliminate duplication and improve the coordination of services. This can be done in a number of ways, including the merging of different departments or agencies, and the introduction of new management practices.

Finally, it is also important to introduce new management practices, which can help to improve the efficiency of the public sector. This can be done in a number of ways, including the introduction of new performance measures, and the use of new technologies.

There are a number of challenges facing the public sector, and it is important to find ways to overcome them. This requires a combination of different approaches, and it is important to continue to work on improving the efficiency of the public sector.

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There are a number of challenges facing the public sector, and it is important to find ways to overcome them. This requires a combination of different approaches, and it is important to continue to work on improving the efficiency of the public sector.

SEMESTERWISE CURRICULUM STRUCTURE

Computer Engineering Program: Semester 4

Course Code	Course Name	Teaching Scheme (Contact Hours/Week)			Credits Assigned				Examination Scheme						
		L	T	P	L	T	P	Total	Theory Courses				Practical / Lab		Total Marks
									ISE 1	ISE 2	MSE	ESE	ISE 1	ISE 2	
CSC401	Design and Analysis of Algorithms	3	1	0	3	1	0	4	20	20	-	60	-	-	100
CSL401	Algorithm Design Laboratory	0	0	2	0	0	1	1	20	20	-	-	25	25	100
CSC402	Software Engineering	3	0	0	3	0	0	3	20	20	-	60	-	-	100
CSC403	Computer Architecture	3	1	0	3	1	0	4	20	20	-	60	-	-	100
CSE404	Professional Elective: Cyber Security	3	0	0	3	0	0	3	20	20	-	60	-	-	100
CSO405	Open Elective: Data Visualization	2	0	2	2	0	1	3	20	20	-	60	25	25	150
CSP406	Community Innovation Project	0	0	4	0	0	2	2	20	20	-	-	25	25	100
Total		14	2	8	20				-						

Abbreviations: L: Lecture; T: Tutorial; P: Practical/Lab; ISE: In-Semester Evaluation; MSE: Mid-Semester Examination; ESE: End Semester Examination.

Regulatory Ordinance Code: R-2026-COMP-IV

the 1990s, the number of people in the UK who are employed in the public sector has increased by 1.5 million, from 2.5 million in 1980 to 4 million in 1999 (Department of Health 2000).

There is a growing emphasis on the need to improve the quality of care in the public sector, and to ensure that the public sector is able to meet the needs of the population. This has led to a number of initiatives, including the introduction of the Health Care Act 1999, which aims to improve the quality of care in the public sector, and the introduction of the Health Care Act 2001, which aims to improve the quality of care in the public sector.

The Health Care Act 1999 introduced a number of measures to improve the quality of care in the public sector, including the introduction of the Health Care Act 1999, which aims to improve the quality of care in the public sector, and the introduction of the Health Care Act 2001, which aims to improve the quality of care in the public sector.

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the 1990s, the number of people in the world who are obese has increased by 100% (World Health Organization 1997). The prevalence of obesity in the United States has increased from 15% in 1980 to 23% in 1994 (Flegal et al. 1994).

Obesity is a risk factor for a number of chronic diseases, including coronary heart disease, stroke, hypertension, type 2 diabetes, and certain types of cancer (World Health Organization 1997). Obesity is also associated with a number of psychological problems, including depression, anxiety, and low self-esteem (Flegal et al. 1994).

Obesity is a complex condition, and its causes are not fully understood. It is thought to be caused by a combination of genetic, environmental, and behavioral factors. Genetic factors may include a predisposition to obesity, which is inherited from one or both parents. Environmental factors may include a diet high in calories and fat, and a lack of physical activity. Behavioral factors may include a tendency to eat large portions, and a tendency to be sedentary.

Obesity is a serious public health problem, and it is important to understand its causes and consequences. This paper will review the current research on the causes and consequences of obesity, and will discuss some of the strategies that have been developed to prevent and treat obesity.

The first section of the paper will discuss the prevalence of obesity in the United States and other developed countries. The second section will discuss the health consequences of obesity, including the risk of chronic diseases and psychological problems. The third section will discuss the causes of obesity, including genetic, environmental, and behavioral factors. The fourth section will discuss some of the strategies that have been developed to prevent and treat obesity.

The prevalence of obesity in the United States has increased steadily since the 1980s. In 1980, 15% of the adult population was obese, and by 1994, this figure had risen to 23%. The increase in obesity has been particularly pronounced among children and adolescents. In 1980, 7% of children and adolescents were obese, and by 1994, this figure had risen to 12%.

The health consequences of obesity are serious. Obesity is a risk factor for a number of chronic diseases, including coronary heart disease, stroke, hypertension, type 2 diabetes, and certain types of cancer. Obesity is also associated with a number of psychological problems, including depression, anxiety, and low self-esteem.

The causes of obesity are complex, and it is thought to be caused by a combination of genetic, environmental, and behavioral factors. Genetic factors may include a predisposition to obesity, which is inherited from one or both parents. Environmental factors may include a diet high in calories and fat, and a lack of physical activity. Behavioral factors may include a tendency to eat large portions, and a tendency to be sedentary.

There are a number of strategies that have been developed to prevent and treat obesity. These include diet and exercise programs, behavioral therapy, and surgery. Diet and exercise programs are the most common, and they can be effective in helping people lose weight. Behavioral therapy can help people develop healthier eating and exercise habits. Surgery is a more radical option, and it is only used in severe cases of obesity.



Society of St. Francis Xavier, Pilar's

Fr. Conceicao Rodrigues College of Engineering

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(Autonomous College affiliated to University of Mumbai)

Course Content

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		3	1	--	--	3	1	--	4
CSC301	Data Structures and Algorithms	Examination Scheme							
			ISE1	MSE	ISE2	ESE	Total		
		Theory	20	--	20	60	100		

Pre-requisite Course Codes	Engineering mathematics, programming fundamentals, and discipline-specific foundation courses.
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify data structures and algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain data structures and algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply data structures and algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze data structures and algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design data structures and algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total				46

Module No.	Unit No.	Topics	Ref.	Hrs.
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			46	

Course Assessment:

Theory:

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Data Structures and Algorithms: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Data Structures and Algorithms. Pearson, 1st, 2024.

Department Faculty Board. Data Structures and Algorithms Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



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Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

(Autonomous College affiliated to University of Mumbai)

Course Content (Practical Only)

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		--	--	2	--	--	--	1	1
		Examination Scheme							
CSL301	Data Structures Laboratory		ISE1	MSE	ISE2	ESE	Total		
		Theory	--	--	--	--	--		
		Lab	25	--	25	--	50		

Pre-requisite Course Codes	Nil
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify data structures laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain data structures laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply data structures laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze data structures laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design data structures laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Course Assessment: (Lab)

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

Laboratory Rubric (25 marks): Continuous laboratory evaluation using predefined rubrics and viva voce.

To be Taught in laboratory			
	Topics wise List of Experiments with relevant topic	Ref.	COs
1	Data Structures Laboratory Experiment 1: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
2	Data Structures Laboratory Experiment 2: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
3	Data Structures Laboratory Experiment 3: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
4	Data Structures Laboratory Experiment 4: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
5	Data Structures Laboratory Experiment 5: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
6	Data Structures Laboratory Experiment 6: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--

Recommended Books:

A. Kulkarni, M. Rao. Data Structures Laboratory: Principles and Practice. University Press, 2nd, 2025.
S. Mehta. Advanced Topics in Data Structures Laboratory. Pearson, 1st, 2024.
Department Faculty Board. Data Structures Laboratory Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



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Course Content

Course Code CSC302	Course Name Database Management Systems	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		3	--	--	--	3	--	--	3
		Examination Scheme							
			ISE1	MSE	ISE2	ESE	Total		
		Theory	20	--	20	60	100		

Pre-requisite Course Codes	Engineering mathematics, programming fundamentals, and discipline-specific foundation courses.
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify database management systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain database management systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply database management systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze database management systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design database management systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			46	

Module No.	Unit No.	Topics	Ref.	Hrs.
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
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	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			46	

Course Assessment:

Theory:

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

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Department Faculty Board. Database Management Systems Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



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Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

(Autonomous College affiliated to University of Mumbai)

Course Content (Practical Only)

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		--	--	2	--	--	--	1	1
CSL302	Database Systems Laboratory	Examination Scheme							
			ISE1	MSE	ISE2	ESE	Total		
		Theory	--	--	--	--	--		
		Lab	25	--	25	--	50		

Pre-requisite Course Codes	Nil
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify database systems laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain database systems laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply database systems laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze database systems laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design database systems laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Course Assessment: (Lab)

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

Laboratory Rubric (25 marks): Continuous laboratory evaluation using predefined rubrics and viva voce.

To be Taught in laboratory			
	Topics wise List of Experiments with relevant topic	Ref.	COs
1	Database Systems Laboratory Experiment 1: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
2	Database Systems Laboratory Experiment 2: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
3	Database Systems Laboratory Experiment 3: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
4	Database Systems Laboratory Experiment 4: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
5	Database Systems Laboratory Experiment 5: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
6	Database Systems Laboratory Experiment 6: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--

Recommended Books:

A. Kulkarni, M. Rao. Database Systems Laboratory: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Database Systems Laboratory. Pearson, 1st, 2024.

Department Faculty Board. Database Systems Laboratory Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



Society of St. Francis Xavier, Pilar's

Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

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Course Content

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		3	1	--	--	3	1	--	4
CSC303	Computer Networks	Examination Scheme							
			ISE1	MSE	ISE2	ESE	Total		
		Theory	20	--	20	60	100		

Pre-requisite Course Codes	Engineering mathematics, programming fundamentals, and discipline-specific foundation courses.
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify computer networks concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain computer networks concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply computer networks concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze computer networks concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design computer networks concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total				46

Module No.	Unit No.	Topics	Ref.	Hrs.
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			46	

Course Assessment:

Theory:

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Computer Networks: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Computer Networks. Pearson, 1st, 2024.

Department Faculty Board. Computer Networks Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



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Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

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Course Content

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		3	--	--	--	3	--	--	3
CSC304	Operating Systems	Examination Scheme							
			ISE1	MSE	ISE2	ESE	Total		
		Theory	20	--	20	60	100		

Pre-requisite Course Codes	Nil
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify operating systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain operating systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply operating systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze operating systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design operating systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			46	

Module No.	Unit No.	Topics	Ref.	Hrs.
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total				46

Course Assessment:

Theory:

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Operating Systems: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Operating Systems. Pearson, 1st, 2024.

Department Faculty Board. Operating Systems Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



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Course Content

Course Code CSE305	Course Name Professional Elective: Artificial Intelligence	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		3	--	--	--	3	--	--	3
		Examination Scheme							
			ISE1	MSE	ISE2	ESE	Total		
		Theory	20	--	20	60	100		

Pre-requisite Course Codes	Engineering mathematics, programming fundamentals, and discipline-specific foundation courses.
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify professional elective: artificial intelligence concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain professional elective: artificial intelligence concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply professional elective: artificial intelligence concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze professional elective: artificial intelligence concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design professional elective: artificial intelligence concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			46	

Module No.	Unit No.	Topics	Ref.	Hrs.
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			46	

Course Assessment:

Theory:

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Professional Elective: Artificial Intelligence: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Professional Elective: Artificial Intelligence. Pearson, 1st, 2024.

Department Faculty Board. Professional Elective: Artificial Intelligence Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



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Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

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Course Content (includes Practical)

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		--	--	4	--	--	--	2	2
		Examination Scheme							
CSP306	Mini Project I		ISE1	MSE	ISE2	ESE	Total		
		Theory	20	--	20	--	50		
		Lab	25	--	25	--	50		

Pre-requisite Course Codes	Nil
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify mini project i concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain mini project i concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply mini project i concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze mini project i concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design mini project i concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total				0

Module No.	Unit No.	Topics	Ref.	Hrs.
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			0	

Course Assessment:

Theory:

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

Laboratory Rubric (25 marks): Continuous laboratory evaluation using predefined rubrics and viva voce.

To be Taught in laboratory			
	Topics	Ref.	COs
1	Mini Project I Experiment 1: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
2	Mini Project I Experiment 2: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
3	Mini Project I Experiment 3: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
4	Mini Project I Experiment 4: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
5	Mini Project I Experiment 5: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
6	Mini Project I Experiment 6: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--

Recommended Books:

A. Kulkarni, M. Rao. Mini Project I: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Mini Project I. Pearson, 1st, 2024.

Department Faculty Board. Mini Project I Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



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Course Content

Course Code CSC401	Course Name Design and Analysis of Algorithms	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		3	1	--	--	3	1	--	4
		Examination Scheme							
			ISE1	MSE	ISE2	ESE	Total		
		Theory	20	--	20	60	100		

Pre-requisite Course Codes	Engineering mathematics, programming fundamentals, and discipline-specific foundation courses.
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify design and analysis of algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain design and analysis of algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply design and analysis of algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze design and analysis of algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design design and analysis of algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			46	

Module No.	Unit No.	Topics	Ref.	Hrs.
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			46	

Course Assessment:

Theory:

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Design and Analysis of Algorithms: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Design and Analysis of Algorithms. Pearson, 1st, 2024.

Department Faculty Board. Design and Analysis of Algorithms Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



Society of St. Francis Xavier, Pilar's

Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

(Autonomous College affiliated to University of Mumbai)

Course Content (Practical Only)

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		--	--	2	--	--	--	1	1
CSL401	Algorithm Design Laboratory	Examination Scheme							
			ISE1	MSE	ISE2	ESE	Total		
		Theory	--	--	--	--	--		
		Lab	25	--	25	--	50		

Pre-requisite Course Codes	Nil
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify algorithm design laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain algorithm design laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply algorithm design laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze algorithm design laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design algorithm design laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Course Assessment: (Lab)

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

Laboratory Rubric (25 marks): Continuous laboratory evaluation using predefined rubrics and viva voce.

To be Taught in laboratory			
	Topics wise List of Experiments with relevant topic	Ref.	COs
1	Algorithm Design Laboratory Experiment 1: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
2	Algorithm Design Laboratory Experiment 2: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
3	Algorithm Design Laboratory Experiment 3: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
4	Algorithm Design Laboratory Experiment 4: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
5	Algorithm Design Laboratory Experiment 5: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
6	Algorithm Design Laboratory Experiment 6: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--

Recommended Books:

A. Kulkarni, M. Rao. Algorithm Design Laboratory: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Algorithm Design Laboratory. Pearson, 1st, 2024.

Department Faculty Board. Algorithm Design Laboratory Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



Society of St. Francis Xavier, Pilar's

Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

(Autonomous College affiliated to University of Mumbai)

Course Content

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		3	--	--	--	3	--	--	3
CSC402	Software Engineering	Examination Scheme							
			ISE1	MSE	ISE2	ESE	Total		
		Theory	20	--	20	60	100		

Pre-requisite Course Codes	Engineering mathematics, programming fundamentals, and discipline-specific foundation courses.
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify software engineering concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain software engineering concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply software engineering concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze software engineering concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design software engineering concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total				46

Module No.	Unit No.	Topics	Ref.	Hrs.
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			46	

Course Assessment:

Theory:

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Software Engineering: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Software Engineering. Pearson, 1st, 2024.

Department Faculty Board. Software Engineering Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



Society of St. Francis Xavier, Pilar's

Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

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Course Content

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		3	1	--	--	3	1	--	4
CSC403	Computer Architecture	Examination Scheme							
			ISE1	MSE	ISE2	ESE	Total		
		Theory	20	--	20	60	100		

Pre-requisite Course Codes	Nil
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify computer architecture concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain computer architecture concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply computer architecture concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze computer architecture concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design computer architecture concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			46	

Module No.	Unit No.	Topics	Ref.	Hrs.
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			46	

Course Assessment:

Theory:

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Computer Architecture: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Computer Architecture. Pearson, 1st, 2024.

Department Faculty Board. Computer Architecture Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



Society of St. Francis Xavier, Pilar's

Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

(Autonomous College affiliated to University of Mumbai)

Course Content

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		3	--	--	--	3	--	--	3
		Examination Scheme							
CSE404	Professional Elective: Cyber Security		ISE1	MSE	ISE2	ESE	Total		
		Theory	20	--	20	60	100		

Pre-requisite Course Codes	Engineering mathematics, programming fundamentals, and discipline-specific foundation courses.
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify professional elective: cyber security concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain professional elective: cyber security concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply professional elective: cyber security concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze professional elective: cyber security concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design professional elective: cyber security concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total				46

Module No.	Unit No.	Topics	Ref.	Hrs.
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			46	

Course Assessment:

Theory:

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Professional Elective: Cyber Security: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Professional Elective: Cyber Security. Pearson, 1st, 2024.

Department Faculty Board. Professional Elective: Cyber Security Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



Society of St. Francis Xavier, Pilar's

Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

(Autonomous College affiliated to University of Mumbai)

Course Content (includes Practical)

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		2	--	2	--	2	--	1	3
CSO405	Open Elective: Data Visualization	Examination Scheme							
			ISE1	MSE	ISE2	ESE	Total		
		Theory	20	--	20	60	100		
		Lab	25	--	25	--	50		

Pre-requisite Course Codes		Nil
After the successful completion students should be able to:		
Course Outcomes	CO1 Identify open elective: data visualization concepts in engineering situations with appropriate evidence, constraints, and validation criteria.	
	CO2 Explain open elective: data visualization concepts in engineering situations with appropriate evidence, constraints, and validation criteria.	
	CO3 Apply open elective: data visualization concepts in engineering situations with appropriate evidence, constraints, and validation criteria.	
	CO4 Analyze open elective: data visualization concepts in engineering situations with appropriate evidence, constraints, and validation criteria.	
	CO5 Design open elective: data visualization concepts in engineering situations with appropriate evidence, constraints, and validation criteria.	

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			46	

Module No.	Unit No.	Topics	Ref.	Hrs.
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			46	

Course Assessment:

Theory:

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

To be Taught in laboratory			
	Topics	Ref.	COs
1	Open Elective: Data Visualization Experiment 1: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
2	Open Elective: Data Visualization Experiment 2: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
3	Open Elective: Data Visualization Experiment 3: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
4	Open Elective: Data Visualization Experiment 4: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
5	Open Elective: Data Visualization Experiment 5: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
6	Open Elective: Data Visualization Experiment 6: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--

Recommended Books:

A. Kulkarni, M. Rao. Open Elective: Data Visualization: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Open Elective: Data Visualization. Pearson, 1st, 2024.

Department Faculty Board. Open Elective: Data Visualization Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3



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Course Content (includes Practical)

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
		--	--	4	--	--	--	2	2
		Examination Scheme							
CSP406	Community Innovation Project		ISE1	MSE	ISE2	ESE	Total		
		Theory	20	--	20	--	50		
		Lab	25	--	25	--	50		

Pre-requisite Course Codes	Engineering mathematics, programming fundamentals, and discipline-specific foundation courses.
After the successful completion students should be able to:	
Course Outcomes	CO1 Identify community innovation project concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain community innovation project concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply community innovation project concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze community innovation project concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design community innovation project concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			0	

Module No.	Unit No.	Topics	Ref.	Hrs.
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			0	

Course Assessment:

Theory:

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.

ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.

Laboratory Rubric (25 marks): Continuous laboratory evaluation using predefined rubrics and viva voce.

To be Taught in laboratory			
	Topics	Ref.	COs
1	Community Innovation Project Experiment 1: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
2	Community Innovation Project Experiment 2: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
3	Community Innovation Project Experiment 3: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
4	Community Innovation Project Experiment 4: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
5	Community Innovation Project Experiment 5: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
6	Community Innovation Project Experiment 6: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--

Recommended Books:

A. Kulkarni, M. Rao. Community Innovation Project: Principles and Practice. University Press, 2nd, 2025.

S. Mehta. Advanced Topics in Community Innovation Project. Pearson, 1st, 2024.

Department Faculty Board. Community Innovation Project Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3

ANNEXURE A: ACADEMIC POLICIES

1. Evaluation Framework

The academic evaluation uses Continuous Internal Evaluation and End Semester Examination components. Students must secure the minimum passing threshold as defined by autonomous college regulations.

2. Grading System

Under the autonomous framework, grading follows a standard relative system converted to a 10-point CGPA scale.

3. Syllabus Revision Protocol

Major syllabus revisions occur periodically to absorb technological shifts. Minor corrections and elective introductions may occur annually with Board of Studies recommendations and Academic Council approval.

the 1990s, the number of people in the UK who are employed in the public sector has increased by 1.5 million, from 2.5 million in 1980 to 4 million in 1995 (Department of Health 1996).

There is a growing emphasis on the need to improve the quality of care in the public sector. The Department of Health (1996) has set out a number of key objectives for the public sector, including the need to improve the quality of care, to reduce waiting times, to improve the efficiency of the system, and to improve the satisfaction of patients and staff. The Department of Health (1996) has also set out a number of key principles for the public sector, including the need to be patient-centred, to be transparent, to be accountable, and to be efficient.

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REVIEWER COMMENTS SECTION

Course	Section	Comment	Status
CSC301	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Resolved
CSC301	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSL301	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Open
CSL301	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSC302	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Resolved
CSC302	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSL302	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Open
CSL302	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSC303	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Resolved
CSC303	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSC304	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Open
CSC304	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSE305	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Resolved
CSE305	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSP306	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Open
CSP306	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSC401	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Resolved

Course	Section	Comment	Status
CSC401	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSL401	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Open
CSL401	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSC402	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Resolved
CSC402	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSC403	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Open
CSC403	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSE404	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Resolved
CSE404	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSO405	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Open
CSO405	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSP406	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Resolved
CSP406	Modules	Check whether long module blocks remain intact across print page boundaries.	Open